

The Full Positive Triangle in Integer Thue Morse Pressure

Research note prepared for Vladimir Reshetnikov with OpenAI

1 October 2026

Abstract

For every integer moment order $m \geq 2$, all even pressure coefficients strictly between degrees $2m$ and $4m$ are positive. More precisely, both the normalized transfer eigenfunction response $[a^{2r}]h_a(1/2)$ and the pressure coefficient $[t^{2m+2r}]p_m(t/\pi)$ are strictly positive whenever $1 \leq r < m$. This closes the full triangular range preceding the first nonlinear term in the eigenvalue feedback logarithm. The main new ingredient is a coefficientwise comparison of dyadic tangent products. Its proof establishes weak log-majorization through a single crossing of the rearranged tangent sequences. An explicit positive lower bound and exact rational checks accompany the result.

Status. This is an unrefereed research note in ordinary mathematics, not a Lean formalization. It is a separate continuation of the first- and higher-coefficient reports; those artifacts are unchanged. No worldwide priority claim is made. The stronger positivity conjecture for individual eigenbasis coefficients and the signs beyond the stated range remain separate questions.

1 The theorem and its range

For the doubling map $T(x) = 2x \pmod{1}$, let $p_m(c)$ be the pressure of $m \log \cos^2 \pi(x - c)$ in the sum-over-preimages convention. The transfer operator is

$$(\mathcal{L}_{m,c}f)(x) = \sum_{Ty=x} \cos^{2m} \pi(y - c) f(y).$$

Its local dominant eigenvalue is $\rho_m(c) = \exp p_m(c)$. Normalize the analytic eigenfunction by $h_{m,c}(0) = 1$, and write

$$a = \tan t, \quad h_a = h_{m, \arctan(a)/\pi}, \quad R_m = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-2m} = \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}}.$$

Define the response and pressure coefficients

$$H_{m,r} = [a^{2r}]h_a(1/2), \quad E_{m,r} = [t^{2m+2r}]p_m(t/\pi).$$

The source manuscript proves that the degree- $2m$ pressure coefficient vanishes [1]. Earlier companions establish the first positive coefficient, the next one, and the sufficient region $m \geq 3r$ [2, 3]. The present result covers the full range before degree $4m$.

Theorem 1.1. *For all integers $m \geq 2$ and $1 \leq r < m$,*

$$H_{m,r} > 0, \quad E_{m,r} > 0.$$

If $s = m - r$ and

$$g_{m,r} = [a^{2r}] \prod_{j \geq 1} \left(1 + a \tan \frac{\pi}{2^{j+1}} \right)^{2m},$$

then

$$H_{m,r} \geq 2 \left(\frac{2}{\pi} \right)^{2m} g_{m,r} \left[2 - (1 - 2^{-2s}) \zeta(2s) \right] \quad (1)$$

$$\geq 2 \left(\frac{2}{\pi} \right)^{2m} \binom{2m}{2r} \left(2 - \frac{\pi^2}{8} \right) > 0. \quad (2)$$

At fixed m , the positive pressure degrees supplied by this theorem are $2m+2, 2m+4, \dots, 4m-2$. The statement is about coefficients; it does not assert a uniform radius of analyticity as m varies. A later coefficient can be negative: the exact value $E_{2,4} = -35360872/93555$ is recalled in Section 5.

2 Local spectral data and the inverse branch formula

The finite Fourier space

$$\mathcal{E}_m = \text{span}\{e^{2\pi i k x} : 1 - m \leq k \leq m - 1\}$$

is invariant, with matrix entries

$$(A_m(c))_{k,\ell} = 2 \cdot 4^{-m} \binom{2m}{m+2k-\ell} e^{-2\pi i (2k-\ell)c}. \quad (3)$$

An out-of-range binomial coefficient means zero. At $c = 0$, the functions

$$F_j(x) = \sin^{2m}(\pi x) \sum_{n \in \mathbb{Z}} (\pi(x+n))^{-(2m-j)}, \quad 0 \leq j \leq 2m-2,$$

belong to $\mathcal{E}_m$ and satisfy $\mathcal{L}_{m,0} F_j = 2^{-j} F_j$. Membership follows by differentiating the partial-fraction expansion of $\csc^2 \pi x$; splitting the periodization into even and odd integers gives the eigenvalue identity. Their leading terms at zero are $(\pi x)^j$, so they form a basis. In particular $F_0(0) = 1$, $F_0 > 0$, and $F_0(1/2) = R_m$. Analytic perturbation gives the normalized eigenfunction and simple dominant eigenvalue near zero.

These data identify the local pressure as well. Positive comparison with the eigenfunction gives $\mathcal{L}_{m,c}^N 1 \asymp \rho_m(c)^N$. For $W_N(x) = \prod_{j < N} \cos^{2m} \pi(T^j x - c)$, its trigonometric degree is at most $m(2^N - 1)$. The usual L^1 Bernstein inequality and interval variation give

$$2^N \int_0^1 W_N \leq \sum_{I \in \mathcal{D}_N} \sup_I W_N \leq (1 + 2\pi m) 2^N \int_0^1 W_N,$$

where $\mathcal{D}_N$ denotes the dyadic intervals of length 2^{-N} . Since $2^N \int W_N = \int \mathcal{L}_{m,c}^N 1$, the logarithmic growth rate is $\log \rho_m(c)$. This is the local finite-model argument used in [1, 2].

Normalize by the fixed-point weight:

$$(V_a f)(x) = \sum_{Ty=x} (\cos \pi y + a \sin \pi y)^{2m} f(y).$$

Its eigenvalue $\lambda_m(a)$ obeys the exact identities

$$\lambda_m(a) = 1 + a^{2m} h_a(1/2), \quad p_m(t/\pi) = 2m \log \cos t + \log \lambda_m(\tan t). \quad (4)$$

The first follows by evaluating at zero. Reflection gives $h_{-a}(1/2) = h_a(1/2)$.

For a half-integer w , put

$$G_{m,r}(w) = [a^{2r}] \prod_{j \geq 1} \left(1 + a \tan \frac{\pi w}{2^j}\right)^{2m}. \quad (5)$$

No tangent has a pole. Its tail is geometrically summable, so the product converges locally uniformly in a for each fixed w .

Lemma 2.1. *For every integer pair $1 \leq r < m$,*

$$H_{m,r} = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-2m} G_{m,r}(w), \quad (6)$$

with absolute convergence.

Proof. Set $U_N(a, x) = V_a^N 1(x)$. Evaluation at zero gives

$$U_{N+1}(a, 0) = U_N(a, 0) + a^{2m} U_N(a, 1/2),$$

so $U_N(a, 0) = 1 + O(a^{2m})$. For $2r < 2m$, the coefficient of a^{2r} in $U_N(a, 1/2)$ is unchanged on dividing by $U_N(a, 0)$. On a sufficiently small complex disk, spectral projection separates the leading eigenvalue from the rest by a uniform modulus ratio below one. The projection of 1 is nonzero at $a = 0$ and hence nearby. Therefore the normalized iterates converge locally uniformly analytically to $h_a(1/2)$, and Cauchy's coefficient formula gives convergence of these coefficients.

Use centered representatives

$$\mathcal{W}_N = \{w \in \mathbb{Z} + 1/2 : |w| < 2^{N-1}\}.$$

The cosine product telescopes, yielding

$$\begin{aligned} [a^{2r}] U_N(a, 1/2) &= \sum_{w \in \mathcal{W}_N} \left(2^N \sin \frac{\pi w}{2^N}\right)^{-2m} \\ &\quad \cdot [a^{2r}] \prod_{j=1}^N \left(1 + a \tan \frac{\pi w}{2^j}\right)^{2m}. \end{aligned} \quad (7)$$

For all half-integers w ,

$$\sum_{j \geq 1} \left| \tan \frac{\pi w}{2^j} \right| < 10|w|. \quad (8)$$

Indeed, for $w > 0$ take $J = \lceil \log_2(2w) \rceil$. Up to J , distance from poles gives $|\tan(\pi w/2^j)| < 2^{j+1}/\pi$. After J , the argument is at most $\pi/4$, so convexity gives the bound $4w/2^j$. The two sums total less than $(16/\pi + 4)w < 10w$; negative w follows by symmetry.

The absolute coefficient in (7) is consequently at most $(20m|w|)^{2r}/(2r)!$. On $\mathcal{W}_N$, the preceding unperturbed factor is at most $(2|w|)^{-2m}$. This gives a uniform summable majorant $O_{m,r}(|w|^{2r-2m})$, since $r < m$. Dominated convergence proves (6) across the full claimed range. $\square$

3 A tangent product comparison

The following inequality is the new ingredient. Write $[a^r]P \leq [a^r]Q$ for every r when comparing products coefficientwise.

Theorem 3.1. *For every positive odd integer n ,*

$$\prod_{j \geq 1} \left(1 + \frac{a}{n} \left| \tan \frac{n\pi}{2^{j+1}} \right| \right) \leq_{\text{coeff}} \prod_{j \geq 1} \left(1 + a \tan \frac{\pi}{2^{j+1}} \right). \quad (9)$$

More strongly, if

$$\theta_j = \frac{\pi}{2^{j+1}}, \quad b_j = \tan \theta_j, \quad x_j = \frac{|\tan(n\theta_j)|}{n},$$

and $x_j^\downarrow$ is their decreasing rearrangement, then

$$\prod_{j=1}^k x_j^\downarrow \leq \prod_{j=1}^k b_j \quad (k \geq 1). \quad (10)$$

Proof. The case $n = 1$ is equality. Assume $n \geq 3$ and set $L = \lfloor \log_2 n \rfloor$, so $2^L < n < 2^{L+1}$. The tail x_j , $j \geq L + 1$, is strictly decreasing to zero. There are only L earlier terms, all positive, so sorting changes only finitely many positions.

Original prefix products. Include $\theta_0 = \pi/2$ and put

$$s_j = \frac{|\sin(n\theta_j)|}{n \sin \theta_j}.$$

The elementary inequality $|\sin(nu)| \leq n|\sin u|$ gives $0 < s_j \leq 1$, while oddness gives $s_0 = s_1 = 1/n$. The double-angle identity implies

$$\frac{x_j}{b_j} = \frac{s_j^2}{s_{j-1}}, \quad \prod_{j=1}^k \frac{x_j}{b_j} = s_k^2 \prod_{j=2}^{k-1} s_j \leq 1 \quad (k \geq 2).$$

At $k = 1$ the ratio is $1/n$. Thus every original prefix product is already bounded.

The initial sorted terms. We show $x_k^\downarrow \leq b_k$ for $k \leq L$. Oddness of n gives $|\tan(n\theta_j)| \leq \cot \theta_j$: its dyadic grid point stays at least one grid step from every tangent pole. If $j \leq L - k + 1$, then

$$x_j < \frac{2^{j+1}}{\pi n} < \frac{2^{2-k}}{\pi} < \frac{\pi}{2^{k+1}} < b_k,$$

using $8 < \pi^2$. If $j \geq L + 2$, then $n\theta_j < \pi/4$ and

$$x_j < 1/n < 2^{-L} < b_k \quad (k \leq L).$$

Only the k consecutive indices $L - k + 2, \dots, L + 1$ can therefore exceed b_k .

For $k \geq 2$, let $\eta = nb_k$. If $\eta \geq \sqrt{3}$, two adjacent tangent magnitudes cannot both exceed η , because

$$\min\{|\tan u|, |\tan 2u|\} \leq \sqrt{3}.$$

To see this, if $v = |\tan u| > \sqrt{3}$, then $|\tan 2u| = 2v/(v^2 - 1) < \sqrt{3}$. Hence the k candidates cannot all exceed the threshold.

If $\eta < \sqrt{3}$, put $y = n\theta_{L+1}$. Since $k \leq L$,

$$0 < y \leq \frac{1}{2}n\theta_k < \frac{1}{2}\eta < \frac{\sqrt{3}}{2}.$$

On this interval $\tan y < 2y$: convexity on $[0, \pi/3]$ gives $\tan y \leq (3\sqrt{3}/\pi)y < 2y$. Consequently

$$nx_{L+1} = \tan y < 2y \leq n\theta_k < nb_k.$$

Again at most $k - 1$ candidates exceed b_k , proving the initial claim for $2 \leq k \leq L$.

At $k = 1$ only $j = L + 1$ remains to check. Put $q = 2^{L+1} \geq 4$. The function $v \mapsto \tan(cv)/v$ increases for $0 < cv < \pi/2$. Since $n \leq q - 1$,

$$x_{L+1} \leq \frac{\cot(\pi/(2q))}{q-1} < \frac{2q}{\pi(q-1)} \leq \frac{8}{3\pi} < 1 = b_1.$$

Thus $x_k^\downarrow \leq b_k$ throughout $k \leq L$.

The single crossing. For $k \geq L + 1$, each $L + 1 \leq j \leq k$ satisfies

$$x_j \geq x_k = \frac{\tan(n\theta_k)}{n} > \tan \theta_k = b_k,$$

by strict increase of $\tan u/u$ on $(0, \pi/2)$. Every $j > k$ satisfies

$$x_j \leq \frac{4}{\pi}\theta_j \leq \frac{2}{\pi}\theta_k < b_k,$$

using the convex chord bound for tangent on $[0, \pi/4]$. Exactly $k - L$ tail terms exceed b_k , so

$$x_k^\downarrow > b_k \iff \min_{1 \leq j \leq L} x_j > b_k.$$

The right side becomes true once and stays true. Before that crossing, multiplying the individual inequalities gives (10). At and after it, all first k original terms exceed b_k and all later terms are below it. The largest k terms are therefore exactly the first k original terms, whose product was bounded above. This proves (10) using only finite products.

Apply Lemma 3.2 below to the logarithms of the first N sorted terms and to $\log b_1, \dots, \log b_N$. It gives every elementary symmetric coefficient inequality. Both sequences are summable, with geometric tails, so their infinite products converge locally uniformly and their fixed-degree elementary symmetric sums converge monotonically. Letting $N \rightarrow \infty$ proves (9). $\square$

Lemma 3.2. *Let $u, v \in \mathbb{R}^N$ be decreasing vectors with $\sum_{i \leq k} u_i \leq \sum_{i \leq k} v_i$ for every k . If F is differentiable, symmetric, convex, and coordinatewise increasing, then $F(u) \leq F(v)$. In particular, this applies to*

$$F(z) = e_d(e^{z_1}, \dots, e^{z_N}) = \sum_{|I|=d} \exp\left(\sum_{i \in I} z_i\right).$$

Proof. Write $g_i = \partial_i F(u)$. Monotonicity gives $g_i \geq 0$. Symmetry and convexity give $g_1 \geq \dots \geq g_N$: swap two coordinates of u and use monotonicity of the gradient of a convex function; if the coordinates agree, symmetry gives equal derivatives. Set $S_k = \sum_{i \leq k} (v_i - u_i) \geq 0$. Summation by parts yields

$$\nabla F(u) \cdot (v - u) = \sum_{k < N} (g_k - g_{k+1}) S_k + g_N S_N \geq 0.$$

The supporting-plane inequality $F(v) \geq F(u) + \nabla F(u) \cdot (v - u)$ proves the claim. The displayed F is a sum of exponentials of linear functions, so all its required properties hold. $\square$

4 Positivity of the entire triangle

Proof of Theorem 1.1. At $w = 1/2$, all tangents in (5) are positive and the first is 1. Thus $g_{m,r} = G_{m,r}(1/2) \geq \binom{2m}{2r} > 0$. The even coefficient at $w = -1/2$ is equal.

For every positive odd n , take absolute values in the product coefficients and raise (9) to the positive integer power $2m$. Products preserve coefficientwise inequalities with nonnegative coefficients, giving

$$|G_{m,r}(n/2)| \leq n^{2r} g_{m,r}.$$

Pairing the positive and negative half-integers in the absolutely convergent orbit formula therefore yields

$$H_{m,r} \geq 2 \left(\frac{2}{\pi} \right)^{2m} g_{m,r} \left(1 - \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-2(m-r)} \right).$$

The odd-integer sum equals $(1 - 2^{-2s})\zeta(2s) - 1$ at $s = m - r \geq 1$ and decreases with s . At $s = 1$ it is $\pi^2/8 - 1$. This proves (1) and (2).

For pressure, the square in the logarithmic expansion of (4) begins at degree $4m$. Since $r < m$, it contributes nothing at degree $2m + 2r$. The cosine product and feedback identity give exactly

$$E_{m,r} = \sum_{j=0}^r H_{m,j} [t^{2m+2r}] \tan^{2m+2j} t - \frac{m}{m+r} R_{m+r}, \quad H_{m,0} = R_m. \quad (11)$$

All tangent coefficients are nonnegative, and all response coefficients in this range are positive. Moreover, the $j = 0$ term alone dominates the negative term. Indeed,

$$[t^{2r}] \left(\frac{\tan t}{t} \right)^{2m} \geq \frac{\binom{2m}{r}}{3^r} \geq \left(\frac{2m}{3r} \right)^r > \left(\frac{2}{3} \right)^r,$$

while

$$\frac{m}{m+r} \frac{R_{m+r}}{R_m} \leq \frac{m}{m+r} \left(\frac{4}{\pi^2} \right)^r < \left(\frac{4}{\pi^2} \right)^r < \left(\frac{2}{3} \right)^r.$$

The ratio bound for R follows termwise from $|w| \geq 1/2$; the binomial bound follows by multiplying $(2m-i)/(r-i) \geq 2m/r$. The tangent coefficients are positive by $(\tan t)' = 1 + \tan^2 t$. Hence $E_{m,r} > 0$. $\square$

5 Exact checks and remaining boundaries

The package provides rational Fourier recursions independent of the tangent comparison. In the indices of (3), let $j = 2k - \ell$. The normalized matrix has entries

$$(V_a)_{k,\ell} = 2 \cdot 4^{-m} \binom{2m}{m+j} (1 - ia)^{m+j} (1 + ia)^{m-j}.$$

Expand $V_a = \sum_n i^n M_n a^n$, $h_a = \sum_n i^n v_n a^n$, and $\lambda(a) = \sum_n i^n \ell_n a^n$. All matrix coefficients are rational. At each order, compute

$$b_n = \sum_{q=1}^n M_q v_{n-q}, \quad \ell_n = \mathbf{1}^\top b_n,$$

then solve

$$(I - M_0)v_n = b_n - \sum_{q=1}^n \ell_q v_{n-q}, \quad \mathbf{1}^\top v_n = 0.$$

Every residual is checked exactly. Evaluation at $1/2$ uses the signs $(-1)^k$, and $H_{m,r} = (-1)^r \sum_k (-1)^k (v_{2r})_k$. The package verifies all 91 response values with $2 \leq m \leq 14$ and

$1 \leq r < m$, computes their pressure coefficients by (11), and compares the pressure triangle through $m = 8$ against a separate direct phase-eigenvalue recursion. For example,

$$H_{2,1} = 8, \quad H_{3,1} = 155/21, \quad H_{3,2} = 16454/315,$$

$$E_{2,1} = 376/45, \quad E_{3,1} = 213/28.$$

These finite checks corroborate the formulas; the general proof is the tangent comparison and orbit argument.

The stronger conjecture that every even response has nonnegative individual coefficients in the atomic basis F_{2j} is not proved here. Exact exploratory checks through $m = 9$ support it, but the proof of Theorem 1.1 uses only evaluation at $1/2$.

Nor is positivity asserted at every later degree. At $m = 2$, the characteristic equation for the pressure eigenvalue is

$$\chi(\lambda, C) = \lambda^3 - (3/4 + C)\lambda^2 + (1/4 + 5C/8)\lambda - 1/8 = 0, \quad C = \cos 2t.$$

Its branch through $(\lambda, C) = (1, 1)$ is simple, with $\partial_\lambda \chi = 3/8$. Exact substitution gives

$$\begin{aligned} p_2(t/\pi) = & -2t^2 + \frac{376}{45}t^6 + \frac{6836}{189}t^8 + \frac{105448}{2025}t^{10} \\ & - \frac{35360872}{93555}t^{12} + O(t^{14}). \end{aligned}$$

The included standard-library checker certifies this polynomial residual. Determining the general first negative degree after the positive triangle remains open in this note.

References

- [1] *Integer Pressure and a Missing Taylor Coefficient*. ProveIt research manuscript, 28 September 2026. Version inspected 1 October 2026: commit `6a98f89bac85cc6b4ba5d5c3b63996037d9824e1`, blob `1973fd17245864bb8a83e1b1d90416ac032ba7c1`. Versioned repository source.
- [2] *The First Positive Coefficient in Integer Thue Morse Pressure*. Research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Finite Bernoulli response formula, first-coefficient positivity, inverse-branch formula, and large-order expansion.
- [3] *Higher Positive Coefficients in Integer Thue Morse Pressure*. Separate research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Fourth-response positivity, the sufficient region $m \geq 3r$, and the negative degree-twelve certificate.